Series Q Q1

Concepts: what scRNA-seq sees, how it works, how to read the plots

Using a tumour as the worked example, three things made clear in 30 minutes: what is in a tissue, where the data comes from, what the pipeline does.

About 30 min | Prerequisites: none | Example: glioblastoma (GBM) | Deeper: Series A

https://github.com/Charlene717/scRNA-seq-tutorial-Qseries

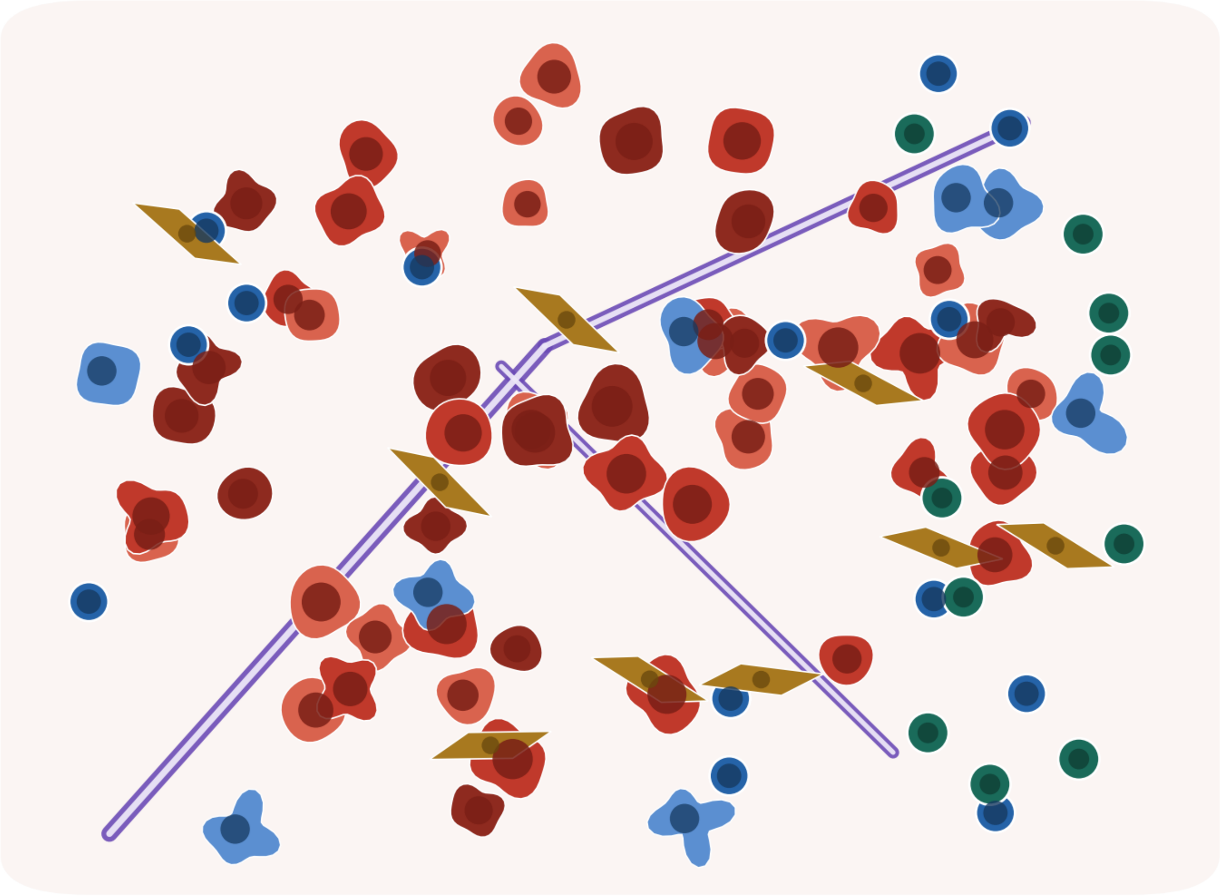
A SCENE

A 58-year-old patient, a 4 cm brain tumour on MRI.

What is actually inside the tissue the surgeon removes?

Pathology says: glioblastoma, grade IV. That name gives you a diagnosis — not the contents of the tissue.

A tumour is an ecosystem, not just malignant cells



one tumour section (schematic)

A tumour is an ecosystem, not a lump of malignant cells

- malignant cells: several states in one tumour
- immune: macrophages/ microglia, T cells
- stroma: fibroblasts, pericytes
- vascular endothelium
- invaded normal tissue (GBM: oligodendrocytes, neurons)

Proportions vary hugely by cancer type, site, even by where you sampled the same tumour—there is no universal set of numbers. That is exactly what single cell measures and bulk cannot read

Malignant cells are only part of it, and come in several states; the rest is immune, stroma, vessels and invaded normal brain

WHERE THIS EPISODE STARTS

Treatment failure, relapse, resistance —
the answer is usually 'which cell, in which state'.

Answering that needs a technology that looks one cell at a time. This episode explains why, and what the data looks like.

Episode goals

1. Name four tumour questions bulk cannot answer, and say why n is patients, not cells

2. Explain how 10x turns a tube of cells into a matrix, and what else sneaks into it

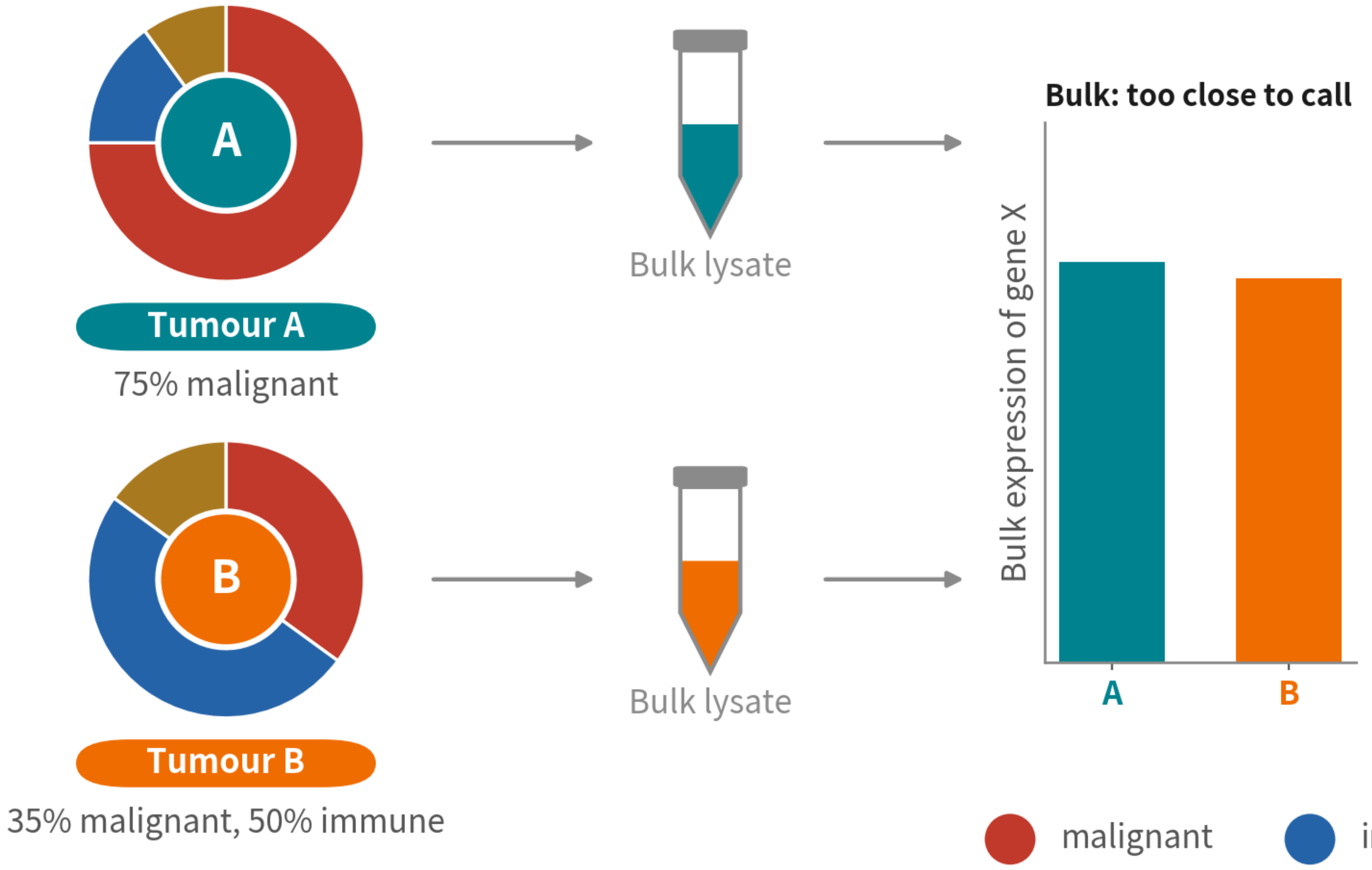
3. Draw the nine steps, name the two tumour-specific phenomena (malignant cells cluster by patient; CNV calls malignancy), and read a UMAP correctly

01

What the traditional method sees, and misses

Understand bulk's limits first; single cell's value then becomes obvious

Bulk RNA-seq: grind the whole tissue, measure one average



Bulk numbers you cannot tell apart, compositions that are nothing alike

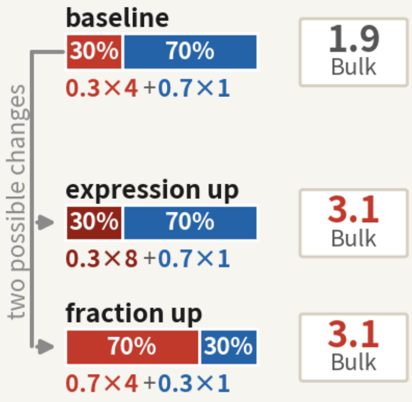
Gene X is high in malignant cells and in immune cells. Bulk sums them, so you cannot tell who produced the signal, or in what proportion

In oncology this is the 'purity' problem: every bulk profile is diluted by stroma and immune cells, by an unknown factor

Two tumours with completely different composition can give bulk numbers too close to tell apart — oncology calls this the purity problem

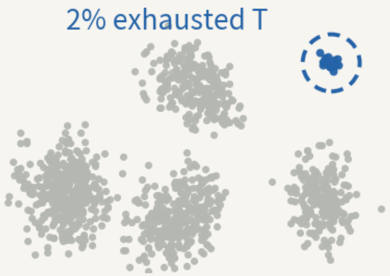
Four tumour questions bulk cannot answer

① Who produced the signal?



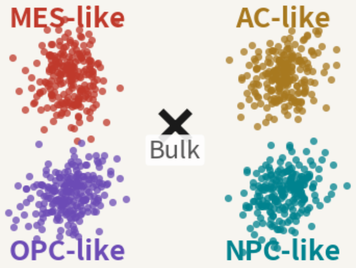
EGFR rises—did each malignant cell change, or did malignant cells get more numerous? Both land on the same bulk value

② What are the rare cells doing?



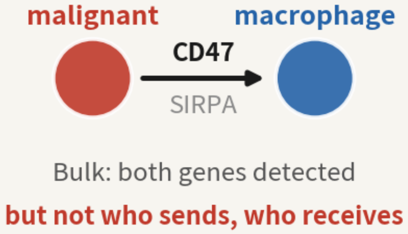
Exhausted T cells or cancer stem cells may be 1–2%. Diluted away in a bulk average; in single cell they form their own cluster

③ How many states do malignant cells have?



GBM malignant cells occupy four states at once and switch between them. Bulk gives you one average position

④ Who is talking to whom?



Ligand and receptor are both detectable in bulk, but not who sends and who receives. Cell communication needs cell-level identity

Who produced the signal, what the rare cells are doing, how many states malignant cells occupy, who is talking to whom — all four are clinically real

When single cell is not actually needed

One test: does your question need to know who each individual cell is? If not, do not use single cell

Skip it | 'Did the whole change?'

Overall change after treatment, prognostic subtyping, cohorts of hundreds. The answer is one tissue-level number — bulk is cheaper, steadier and better powered

Skip it | types known, just counting

You know which types to count and have markers: flow, IHC or bulk deconvolution (Q3) all answer it, across hundreds of samples

You need it | unknown types or continuous states

Rare populations with no marker, types nobody has named, the continuous states among malignant cells, who signals to whom — the answer IS each cell's identity; average it and it is gone

02

From a piece of tissue to a matrix

Three words are enough: GEM, Barcode, UMI

Operating table to sequencer: four steps

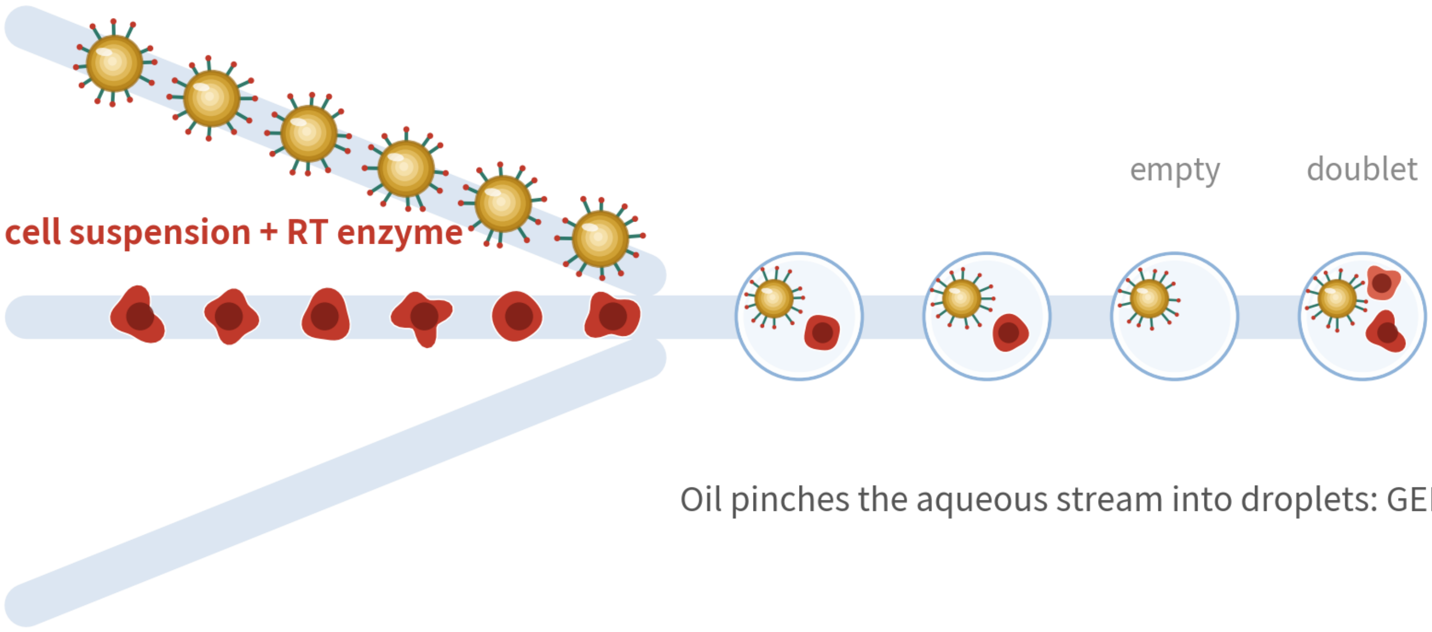
Dissociation	Enzymes break tissue into a single-cell suspension. Tumours are tough and necrotic; this step is brutal and leaves marks
Partitioning	A microfluidic chip seals each cell with a bead in an oil droplet (GEM)
Library	Each cell's mRNA is tagged with its bead's Barcode; everything is pooled into one library
Sequencing & counting	Cell Ranger sorts reads back to cells by Barcode, de-duplicates by UMI, then builds the matrix

More than one platform: 3′, 5′, full-length, nuclei

This series uses two: 10x 3′ in Q2, Smart-seq2 in Q3. The differences set QC and CNV parameters

Platform	What is read	Cells / depth	Best for	Watch out
10x 3′ (Q2)	3′ end of mRNA + UMI	5–10k cells 20–50k reads	Composition, types, states the default	Doublets, ambient RNA
10x 5′ + V(D)J	5′ end + TCR / BCR sequences	as above	Immunotherapy, T / B clonotypes	No extra information on tumour cells
Smart-seq2 (Q3)	Full-length transcripts, no UMI	Hundreds–thousands ~1M reads	CNV, splicing, mutations few precious cells	Costly, slow reads ≠ molecules
snRNA-seq (nuclei)	Nuclear RNA, mostly unspliced	as 10x	Frozen tissue, brain dissociation-fragile tissue	mt ≈ 0 immune cells under-represented

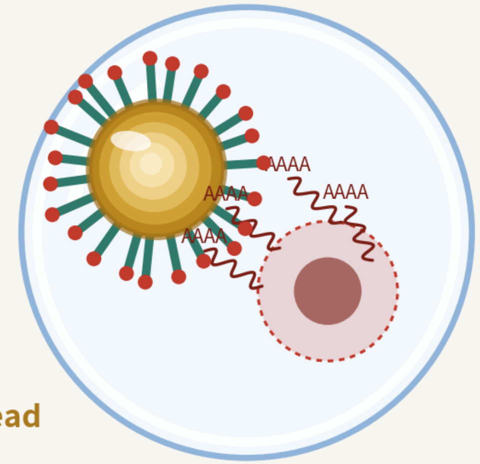
A GEM: one cell, one bead, one droplet



Oil pinches the aqueous stream into droplets: GEMs

Bead–cell pairing is random: most droplets are empty, a few catch two cells (doublets)

Inside one GEM



1 cell+1 bead+ lysis buffer

The cell lyses beside the bead; every mRNA's poly(A) tail is captured on that bead's primers—the bead's barcode becomes the cell's ID

Bead–cell pairing is random: most droplets are empty, a few catch two cells. Both become QC's job later

The Barcode records which cell; the UMI records which molecule

Every primer on a bead

5' | adapter | Cell Barcode (16 nt) | UMI (12 nt) | poly(dT) | 3'

Same bead: identical Barcodes → which cell

ATCGGTAC···	AGTCCA	TTTTTTTT
ATCGGTAC···	TTGACG	TTTTTTTT
ATCGGTAC···	CAGGTA	TTTTTTTT

Every primer a different UMI → which molecule

Why UMIs

Three identical reads:

AGTCCA
AGTCCA
AGTCCA

same UMI→PCR copies of one molecule, count 1

AGTCCA
TTGACG

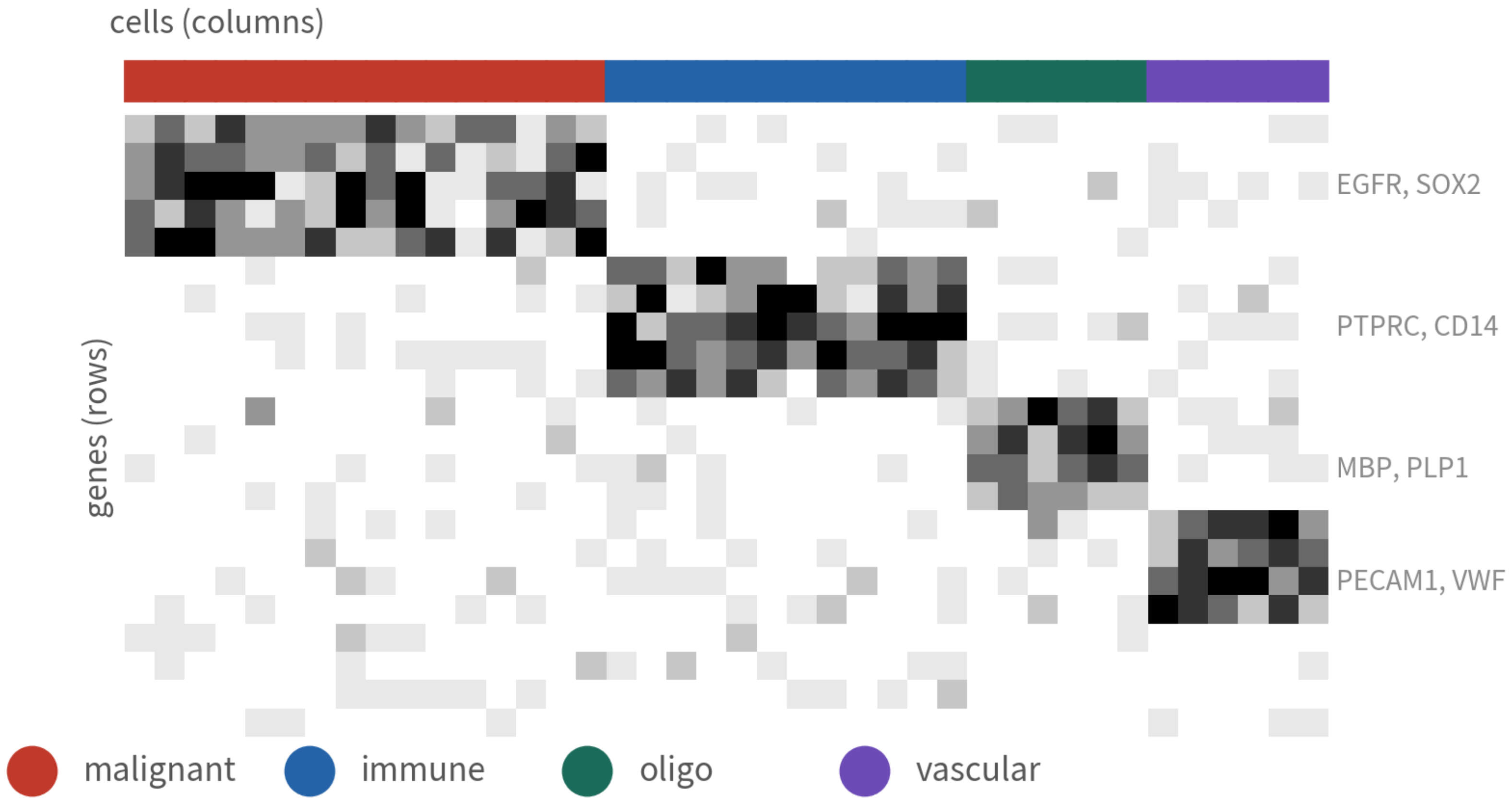
different UMI→two molecules, count 2

So the matrix holds molecule counts (UMIs), not reads
—far smaller than bulk read counts, and more honest

The numbers in the matrix are molecules, not reads — the most fundamental difference from bulk

The output: a gene-by-cell count matrix

What Cell Ranger hands you: a gene × cell count matrix



GBM 5k example: ~36k genes × 5,604 cells; each entry is a UMI count, >90% of entries are zero

Three files

`Barcodes.tsv`
one Barcode per column

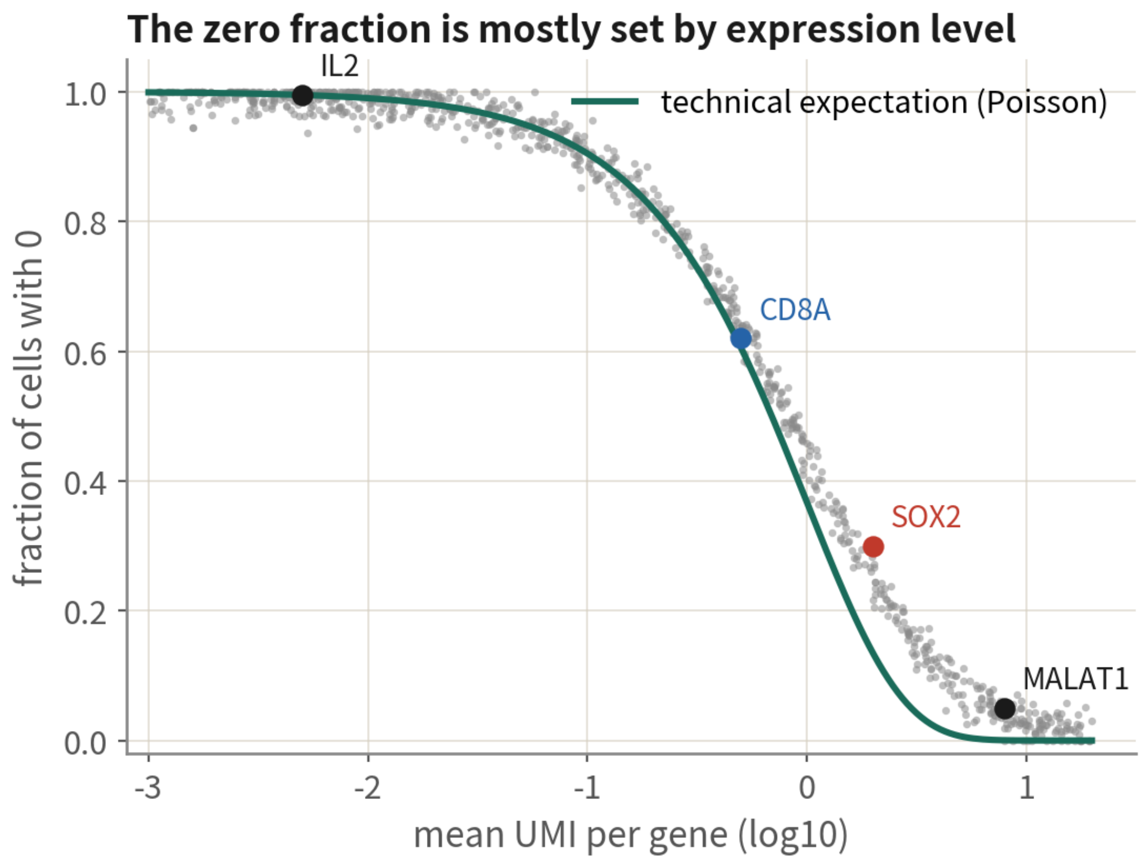
`features.tsv`
one gene per row

`matrix.mtx`
only non-zero entries (sparse)

One column per cell, one row per gene; over 90% zeros — the nature of the technology, not broken data

Ninety percent zeros: two kinds of zero, and conclusions rest on groups of cells

Ninety percent zeros: the nature of the technology, and the starting point of analysis



A zero has two possible meanings

1 **truly not expressed**

e.g. SOX2 in a T cell—this zero is signal, and it is what we want

2 **expressed but not captured (dropout)**

Only 10–30% of a cell's mRNA is captured; low genes are often missed. This zero is noise—recovered by looking at cells as a group

So: never conclude from one gene in one cell; conclusions rest on the distribution across a group of cells

The zero fraction is mostly set by expression level — technical expectation, not broken data. One gene in one cell means nothing; that is why every analysis looks at groups

Not every column is a cell

dying/broken cells

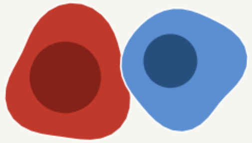
Tumour dissociation is brutal: necrotic regions, enzymatic digestion; percent.mt spikes and is often bimodal

empty droplets (ambient RNA)

Necrosis releases lots of free RNA; empties carry malignant genes and look like low-quality malignant cells

doublets

A malignant cell stuck to a macrophage reads as a 'tumour cell expressing immune genes'—it has happened in the literature

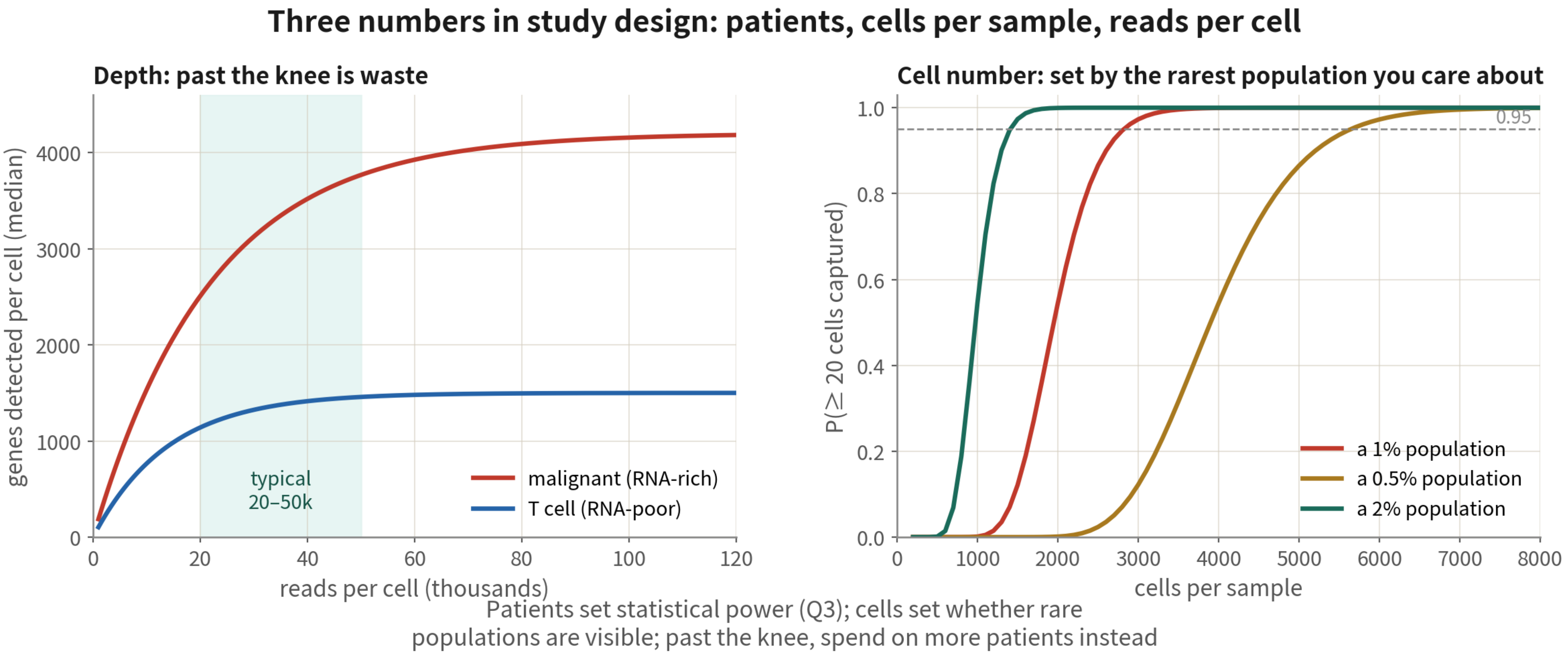


dissociation stress

FOS/JUN/HSPA1A-high cells scattered across types. Not junk, but their 'activation' is an artefact

Tumour samples have plenty of all four. Each has a barcode and looks like a cell — hence QC is step one

Three numbers in study design: patients, cells, depth



Decide in this order: patients (power), then cells (rarest population), then depth (waste past the knee). On a fixed budget, one more patient nearly always beats 10,000 more cells

03

What the pipeline does

Remember the problem each step solves and you have the pipeline; the last page is the red line that runs through all of it

Nine steps, one problem each

The nine standard steps: one problem each

Dark = steps that matter most for tumour data; red cards = what is special about tumours

1	2	3	4	5	6	7	8	9
QC	**Normalize**	**HVG**	**Scale**	**PCA**	**Cluster**	**UMAP**	**Annotate**	**DE**
drop what is not a cell	remove depth	2,000 of 20,000 genes	equalise gene scale	compress to tens of dims	communities in PC space	for the eyes only	name the types	what differs between conditions
bimodal mt, more doublets	malignant cells carry more RNA	patient-specific genes get in		PC1 often= malignant vs normal	malignant cluster by patient		+CNV malignant calling	unit= patient

The standard pipeline ends at DE. Every road after it (malignant calling, communication, trajectory, deconvolution) needs the right data—Q3 gives you the table

This is the route Q2 runs line by line on a real GBM sample. Red cards mark what is special about tumours

After the nine steps: six downstream roads

After the standard pipeline: six downstream roads

The nine steps end with every cell named. The question then picks the road — and the data decides if you can walk it

end of the standard pipeline

every cell now has a name

→

malignant calling

is this cell tumour or normal?

needs trustworthy normal cells as reference

Q3 • 05_infercnv.R

composition & DE

which cell type shifts, and which genes change?

the unit is the patient —you need several

Q3 • 06a / 06b

cell–cell communication

who talks to whom, along which pathway?

both sender and receiver need enough cells

Q3 • 07_cellchat.R

trajectory / pseudotime

which state are these cells moving towards?

needs a real continuum —and you pick the root

Q3 • 08_trajectory.R

pathway & TF activity

are pathways and transcription factors active?

activity is inferred, not measured expression

Q3 • 09_activity.R

deconvolution & survival

does this hold up in a large cohort?

needs an external bulk cohort with outcomes

Q3 • 10_deconv_survival.R

red = precondition

when it fails, say so— don't draw the plot anyway

These are not six menu items but six different questions. The red line on each card is its precondition — when it fails, say so rather than drawing the plot anyway

Analysis happens in PC space; UMAP is only a projection

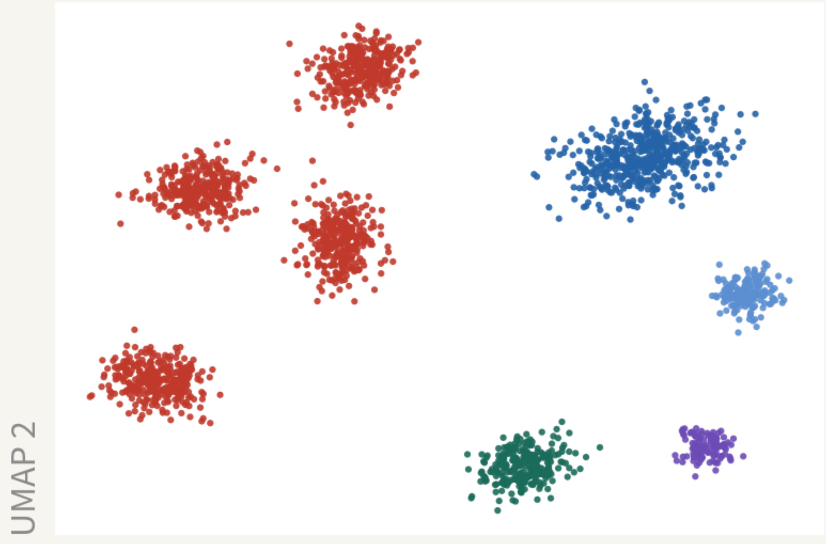
PC space (10–30 dims, invisible)

- clustering: KNN graph + communities
- cell–cell distances, neighbours
- integration (Harmony corrects here)
- trajectory, pseudotime

RunUMAP

projection

UMAP (2 dims, visible)



For looking only: which cells resemble each other. Distance, area and shape are not evidence

Clustering, distances, integration and trajectories all happen on the left; the right is just for your eyes

The red line through the pipeline: the unit is the patient, not the cell

Design 1: one patient, 30,000 cells

30,000

n = 1

statistical unit

Design 2: six patients, 5,000 each

5,000 5,000 5,000 5,000 5,000 5,000

n = 6

supports patient-level inference

30,000 cells from one patient share genome, age, drugs and dissociation — they are not independent observations

30,000 cells from one patient is still n = 1. Testing cells as samples inflates the degrees of freedom into false significance — Q3 measures how big the effect is

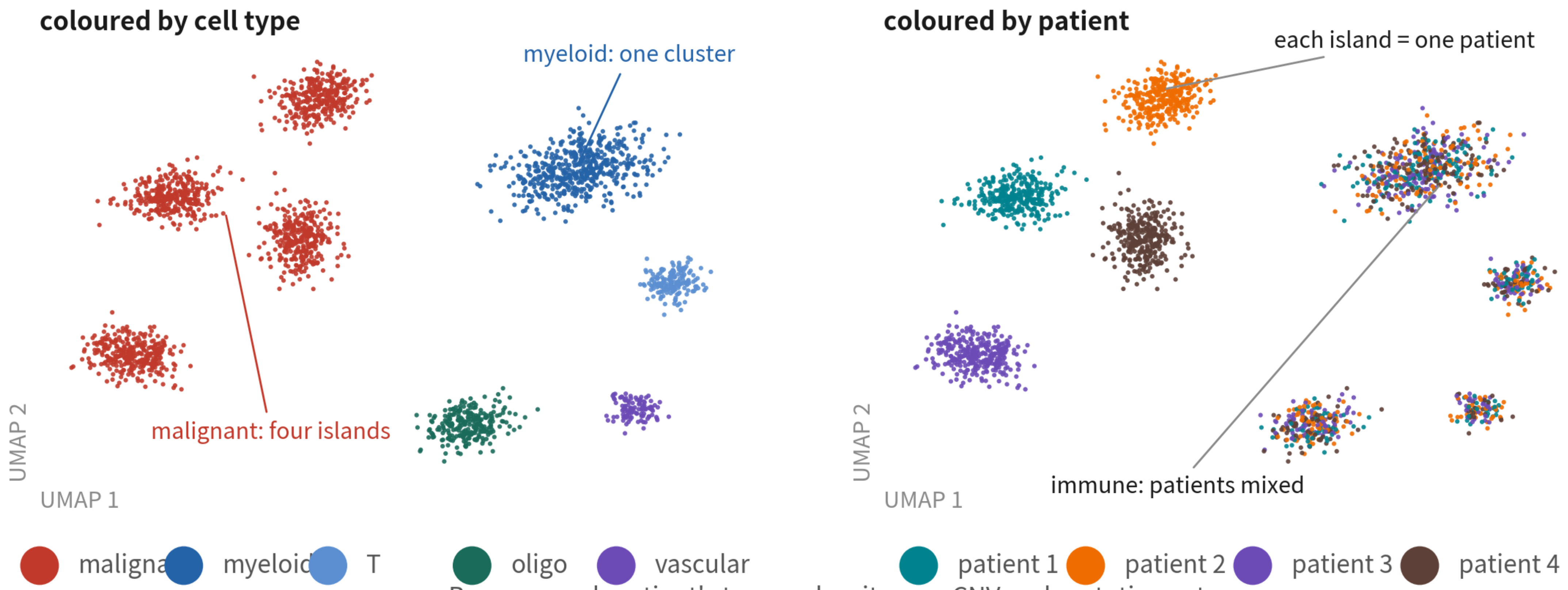
04

What is special about the example: two tumour phenomena

Every tissue has its quirks; these two are where tumour data misleads most

Phenomenon 1: malignant cells cluster by patient, normal cells mix

The defining phenomenon of tumour data: malignant cells cluster by patient, normal cells mix across patients

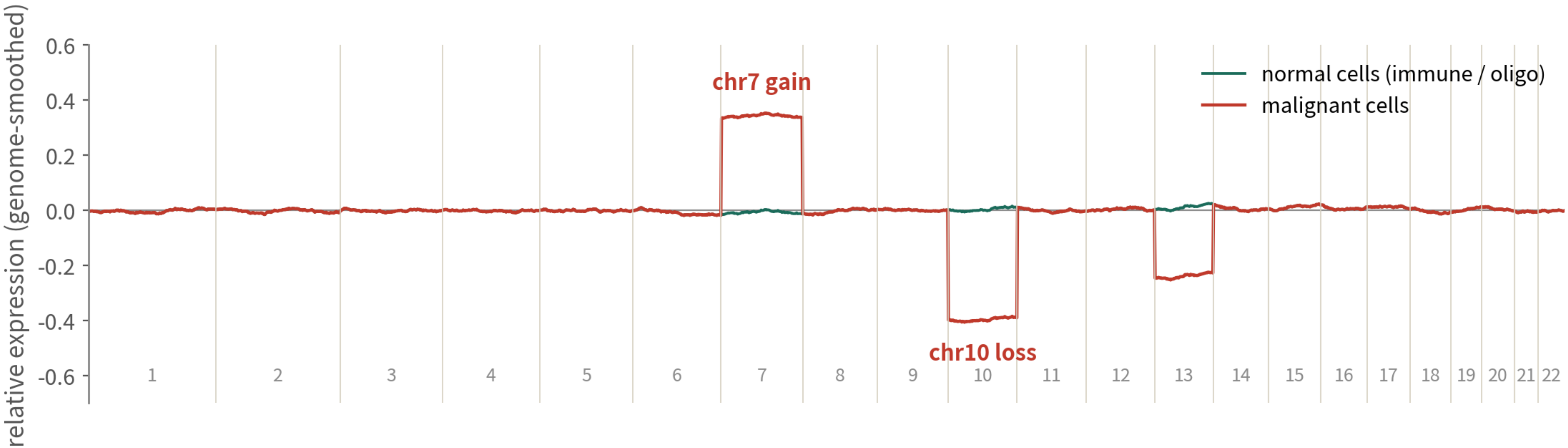


Because each patient's tumour has its own CNV and mutation set.
This is not a batch effect—be careful not to 'correct' it away (Q3)

Each patient's tumour carries its own CNV and mutation set. This is not a batch effect — do not correct it away (Q3)

Phenomenon 2: the markers that would settle it are unreadable; the readable ones cannot settle it

A malignant cell's ID is its genome: infer CNV from chromosome-scale expression trends



One gene says nothing; a hundred neighbouring genes on the same chromosome shifting together has only one explanation—the DNA copy number changed. That is inferCNV, and the most reliable malignant-vs-normal evidence in tumour data (hands-on in Q3)

Mutation markers like IDH1 R132H and H3K27M really do settle malignancy — but they are point mutations, invisible to 3' sequencing. What expression can read, SOX2 and EGFR, only gives lineage. Hence the fallback: a whole chromosome arm shifting together

The four major groups: three are easy, one is not

The first cut in annotating tumour data is always these four

Three: immune · vascular · oligo

Naming ends here

- Immune — PTPRC
- Vascular — PECAM1, VWF
- Oligo — MBP, PLP1 (the inferCNV reference)

The fourth: glial / malignant

Naming is not the end

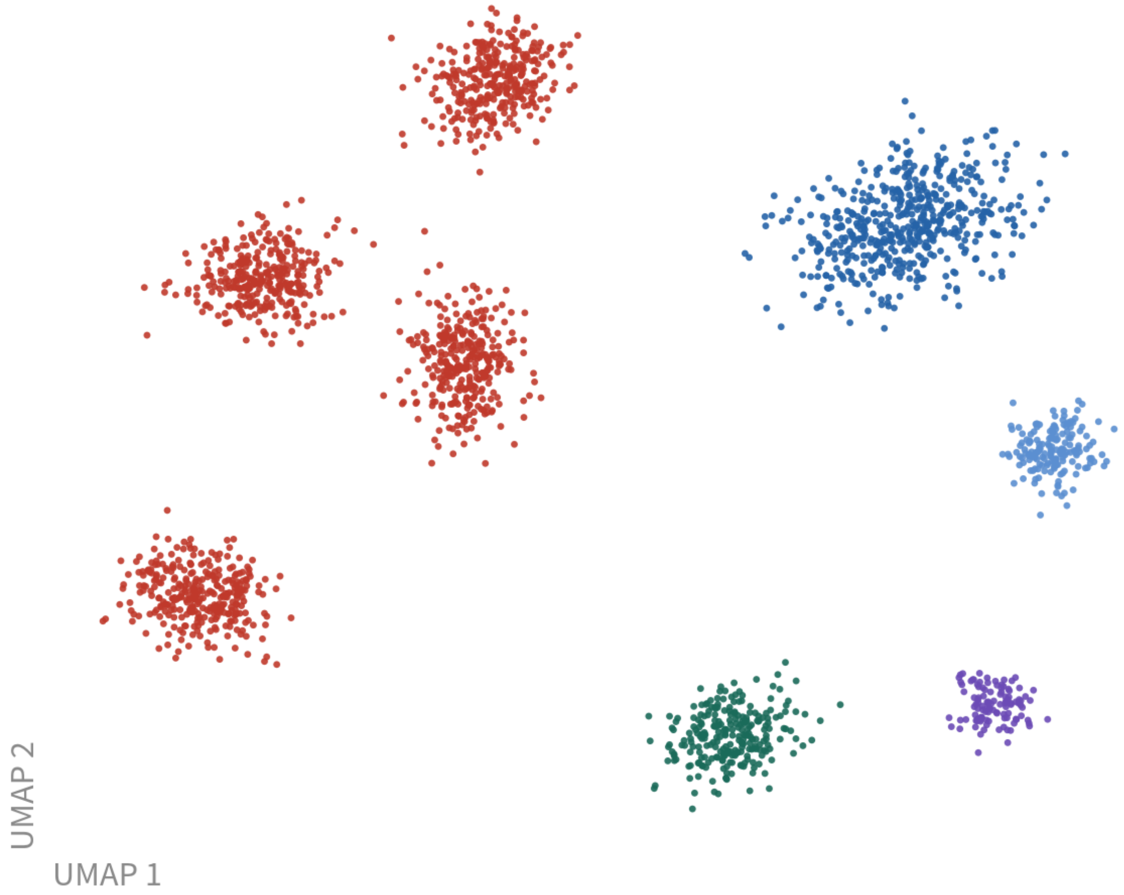
- Glial — SOX2, OLIG2, PTPRZ1, EGFR
- Malignancy needs CNV + patient specificity
- Call first, then states (MES/AC/OPC/NPC)

05

Reading single-cell plots

For most people this is the only skill they will ever need

What a UMAP can and cannot tell us



It can tell you

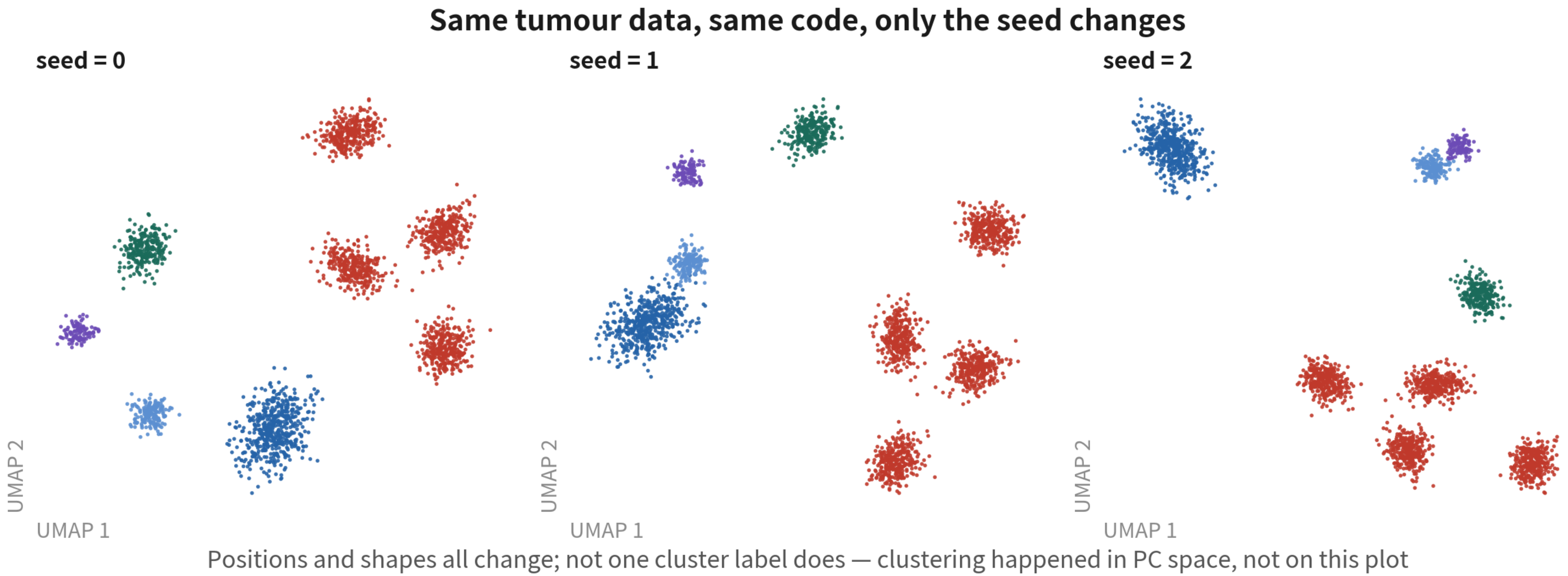
- ✓ which cells resemble each other (local neighbours are reliable)
- ✓ roughly how many groups, which are connected
- ✓ where a gene lights up (FeaturePlot)

It cannot tell you

- ✗ distance ≠ relatedness (a new seed moves it)
- ✗ area ≠ cell count (use table())
- ✗ an elongated shape ≠ a trajectory
- ✗ four malignant islands≠four subtypes (those are four patients)

The last line is tumour-specific: four malignant islands are not four subtypes, they are four patients

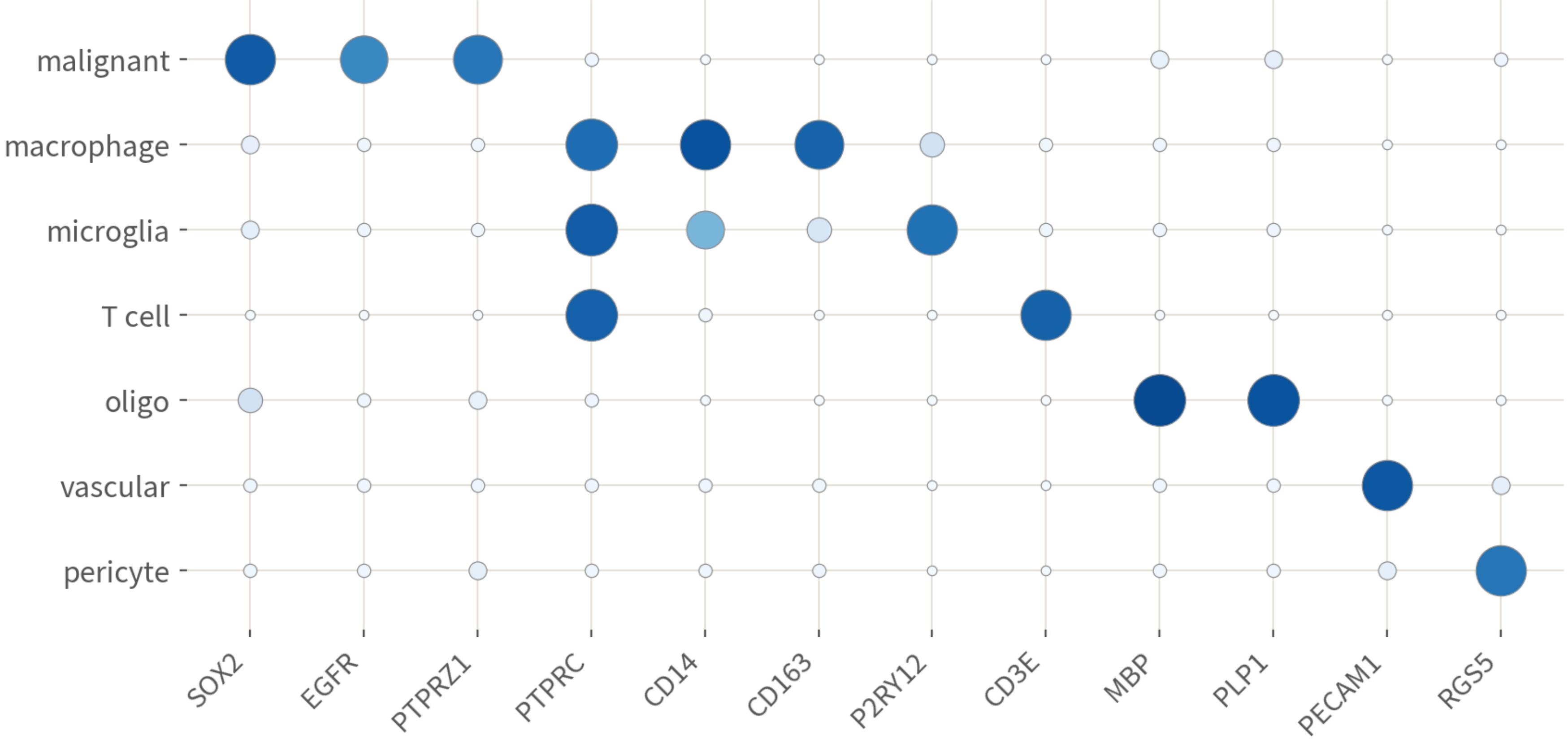
Same data, only the seed changes



Positions and shapes all change, not one label does. 'Cluster A sits near B, suggesting relatedness' carries no evidence

Dotplot: read the two dimensions separately

Dotplot: which group expresses what — a TME marker panel



Read the two dimensions separately

20%

60%

100%

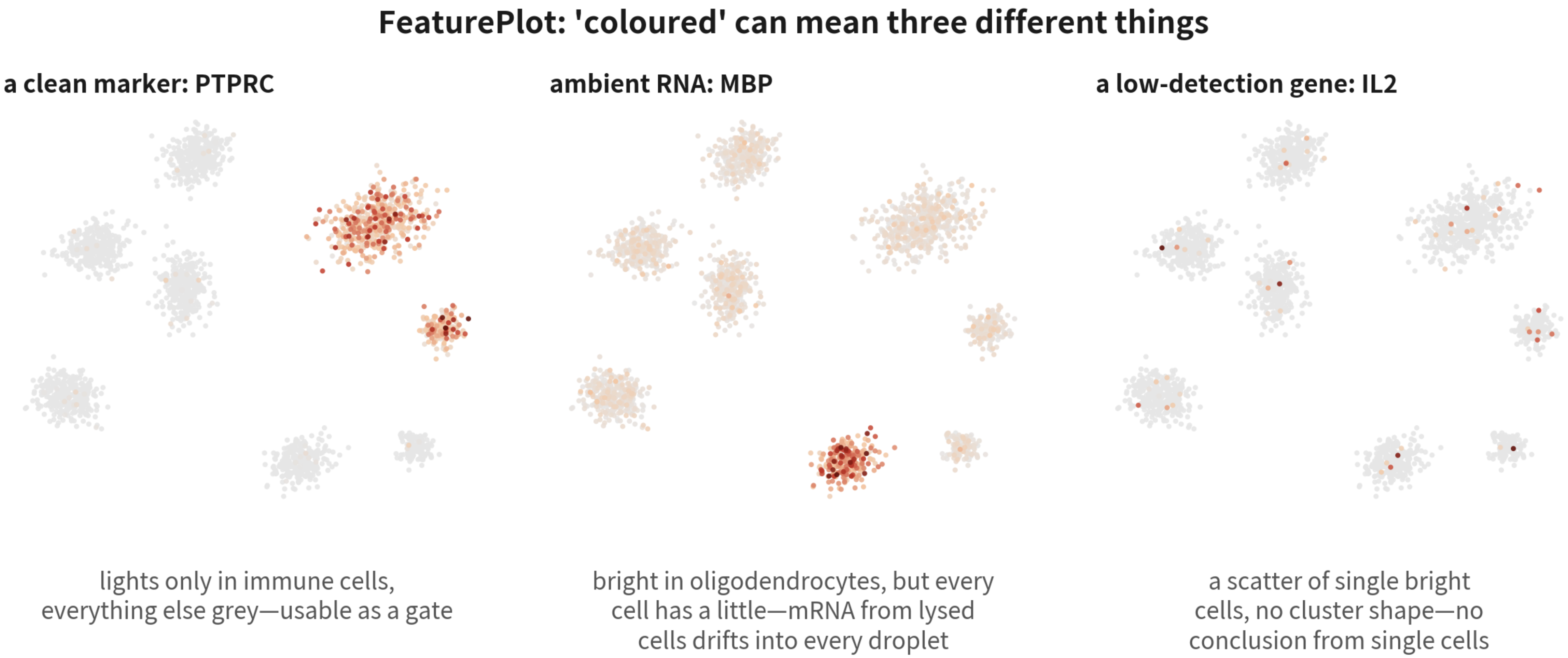
size=fraction of cells expressing

colour = mean expression

Blocks lighting up along the diagonal = good annotation. PTPRC lights three immune types: it marks the immune group, not a type

Size is the fraction of cells expressing, colour the level. Blocks along the diagonal mean a good annotation

FeaturePlot: colour does not always mean expression



A clean gate lights one block; ambient RNA gives every cell a little; a low-detection gene is a sprinkle. First question: does the bright region have the shape of a cluster?

The four most common misreadings

What you see	What you assume	What is true
Two clusters far apart	Biologically distant	UMAP keeps no global distance; a new seed moves them
A big cluster area	Many cells	Area reflects density and parameters; count with table()
A cluster stretched into a band	A trajectory	Could be cell cycle, depth gradient, or a doublet bridge
Malignant cells in four islands	Four malignant subtypes	Four patients; subtypes are read within patients via state scores

Reading Figure 1 of a single-cell paper: five questions first

All five answers live in the methods and supplements; if you cannot find one, discount the conclusion

Question	Why it matters	Where to look
How many patients per group?	n is patients; between-group claims at n < 3 are descriptive	Table 1, 'Samples' in methods
How were cells QC'd and doublets removed?	Without doublet removal, 'dual-lineage' types are usually artefacts	'QC' in methods, Fig S1
What was integrated, what was corrected away?	Integrating by patient flattens real malignant differences (Q3)	'Integration' in methods, pre/post UMAPs
How was malignancy called?	'Tumour cells' by markers alone are not credible; look for CNV or mutations	inferCNV / copyKAT heatmap
Is DE done per cell or per patient?	Cell-level tests hand you thousands of false positives (Q3)	'Differential expression': pseudobulk or not

06

COMMON PITFALLS

Three conceptual errors: no error message, just a wrong conclusion

Pitfall 1: treating cells as samples

What it is

One patient's tumour vs one patient's normal brain, a test over 30,000 cells, 3,000 'disease genes'.

Why it happens

Every cell looks like an independent observation.

What it costs you

Cells from one patient are correlated; n is really 1. Most of those genes will not hold in the next patient.

Pitfall 2: reporting dissociation stress as tumour biology

What it is

A FOS/JUN/HSPA1A-high malignant subset gets written up as a 'stress-response subclone'.

Why it happens

Clean markers, significant enrichment, and it resembles a published 'MES-like state'.

What it costs you

It is the immediate-early response to enzymatic dissociation, present in every type. Tumour tissue dissociates slowly, so the artefact is strong.

Pitfall 3: calling malignancy from markers

What it is

'This cluster expresses EGFR and SOX2, so it is tumour.'

Why it happens

Those genes really are textbook GBM markers.

What it costs you

Reactive astrocytes and OPCs express them too. The evidence is CNV; a 'tumour cell' without CNV support will not survive review.

What this technology cannot see: five limits to state up front

Limits are not flaws, they are the edge of what you may claim — better written by you than by a reviewer

Limit	What it means	What it costs your conclusion
Position is gone	The moment of dissociation, every cell leaves where it used to sit	'Next to a vessel', 'at the edge of necrosis' cannot be said; that needs spatial transcriptomics
RNA only	Protein, modifications and metabolites are not in this matrix	High mRNA is not high protein; 'pathway activation' needs another kind of evidence
A dissociated state	Enzyme, time and temperature all change expression; the FOS/JUN/HSP group comes from exactly this	A 'stress subpopulation' is usually a handling artefact, not tumour biology (see Pitfall 2)
3’ reads the end only	10x 3’ sequences only ~90 bases from each molecule's 3’ end	Point mutations, splicing and fusions are essentially unreadable — hence CNV as the fallback for malignancy
One time point only	Each cell is measured once, and is then gone	'A becomes B' is inference, not observation; a trajectory gives an ordering, not time

What lies beyond single-cell transcriptomes

This series reads RNA only; knowing what else exists tells you what you are missing

Multimodal

RNA + protein / chromatin. CITE-seq adds surface proteins (sharper immune calls); 10x Multiome adds ATAC (regulation, lineage). Same analysis skeleton as this course

Spatial transcriptomics

Visium, Xenium, MERFISH keep position: who sits by vessels, in necrosis, at the core–edge boundary. Single-cell data often serves as the reference for deconvolution

Genome and lineage

TCR / BCR sequencing tracks clones; scDNA-seq or mitochondrial variants track subclones. Tumour-evolution studies pair these with expression

ONE-LINE SUMMARY

A tumour is an ecosystem and single cell takes it apart;
analysis lives in PC space, the UMAP is a projection;
malignancy is read from the genome, and the patient is the unit.

Three sentences. Every line of code in Q2 and every decision in Q3 rests on them.

Check yourself 1

Bulk shows CD8A three-fold higher in tumour A than B. The most conservative reading?

A A's T cells are more cytotoxic

B A may have more CD8 T cells, or each expresses more — Bulk cannot tell

C A will respond better to immunotherapy

Answer B. Composition change and intrinsic change look identical in bulk. A and C over-interpret; you need single-cell data (and multiple patients) to move in that direction.

Check yourself 2

Four GBM patients: malignant cells form four islands, macrophages one blended cluster. This means?

A Severe batch effect; integrate

B Four malignant subtypes

C Expected: malignant cells carry patient-specific CNV, normal cells do not

Answer C. Macrophages blend well, so technical batch is small; malignant islands come from genomic differences. Forcing integration erases real patient biology.

Check yourself 3

The strongest evidence that a cluster is malignant?

A High EGFR and SOX2

B Whole chromosome arms (e.g. chr7 gain, chr10 loss) shifting together

C Sitting far from immune cells on the UMAP

Answer B. Markers only give lineage and UMAP distance is not evidence; CNV is genome-level evidence and the basis of inferCNV.

NEXT

Q2: the single-sample standard pipeline — from raw matrix to cell annotation

Concepts done, now your fingers. Next episode runs 10x's public GBM 5k (5,604 cells) from loading to annotation,
with three practice scripts — every block copy-ready.

https://github.com/Charlene717/scRNA-seq-tutorial-Qseries